

C1 implements I1, I2

1. Aim

To demonstrate how a Java class can implement multiple interfaces using the "implements" keyword.

2. Definition

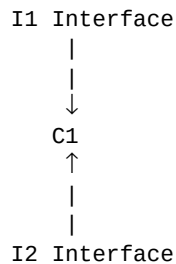
Java allows a class to implement more than one interface. This is used to achieve multiple inheritance through interfaces.

The syntax is:

```
class C1 implements I1, I2
{
    // implementation
}
```

Here, "C1" implements both interfaces "I1" and "I2".

3. Structure



```
interface I1
{
    void method1();
}

interface I2
{
    void method2();
}

class C1 implements I1, I2
{
    public void method1()
    {
        // implementation
    }

    public void method2()
    {
        // implementation
    }
}
```

4. Program

```

interface I1
{
    void displayStudent();
}

interface I2
{
    void displayCollege();
}

public class C1 implements I1, I2
{
    public void displayStudent()
    {
        System.out.println("Student details displayed");
    }

    public void displayCollege()
    {
        System.out.println("College details displayed");
    }

    public static void main(String[] args)
    {
        C1 obj = new C1();

        // Test Case 1
        System.out.println("Test Case 1:");
        obj.displayStudent();

        // Test Case 2
        System.out.println("\nTest Case 2:");
        obj.displayCollege();

        // Test Case 3
        System.out.println("\nTest Case 3:");
        obj.displayStudent();
        obj.displayCollege();
    }
}

```

5. Sample Output

Test Case 1:
Student details displayed

Test Case 2:
College details displayed

Test Case 3:
Student details displayed
College details displayed

6. Test Case Verification

Test Case	Method Called	Expected Output	Result
TC1	"displayStudent()"	Student details displayed	Passed

Test Case	Method Called	Expected Output	Result
TC2	"displayCollege()"	College details displayed	Passed
TC3	"displayStudent()", "displayCollege()"	Both details displayed	Passed

7. Conclusion

Thus, the program successfully demonstrates that class "C1" can implement multiple interfaces "I1" and "I2" using the "implements" keyword. All the methods declared in both interfaces were successfully implemented and executed.